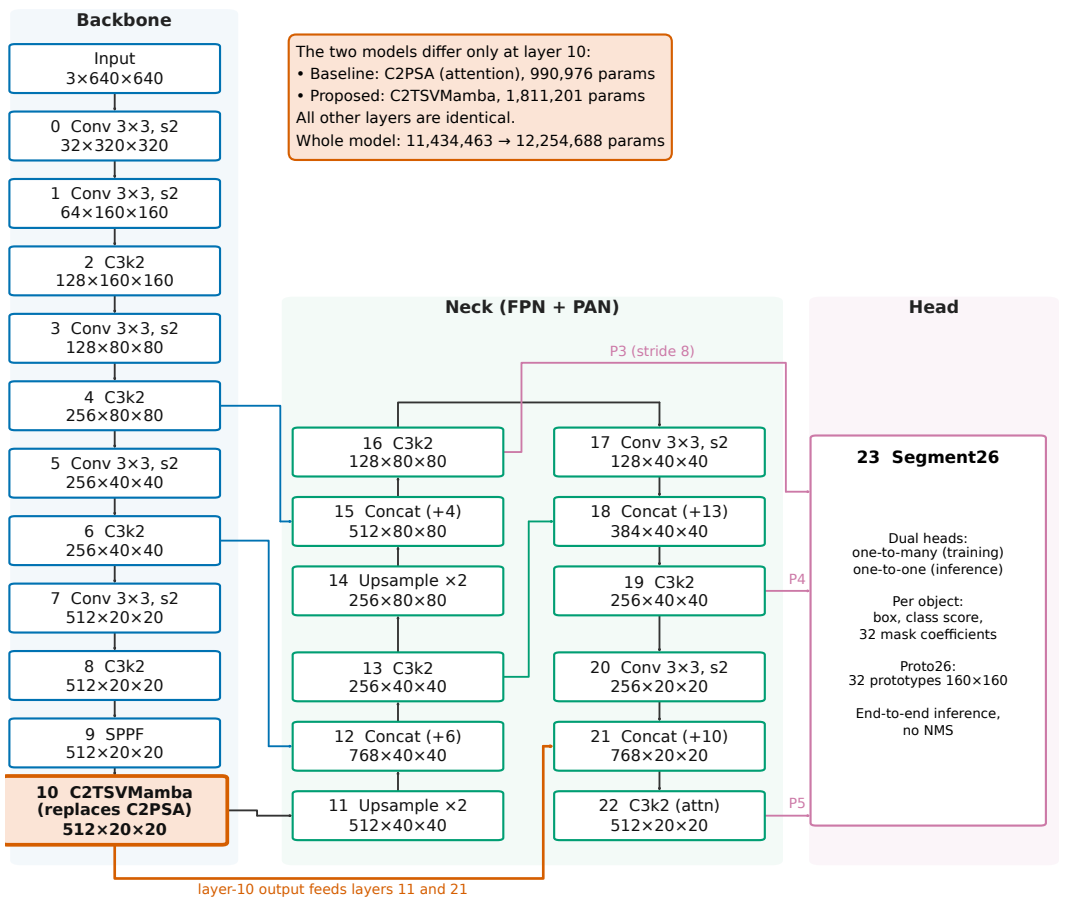
Backbone
Input
3×640×640
0 Conv 3×3, s2
32×320×320
1 Conv 3×3, s2
64×160×160
2 C3k2
128×160×160
3 Conv 3×3, s2
128×80×80
4 C3k2
256×80×80
5 Conv 3×3, s2
256×40×40
6 C3k2
256×40×40
7 Conv 3×3, s2
512×20×20
8 C3k2
512×20×20
9 SPPF
512×20×20
10 C2TSVMamba
(replaces C2PSA)
512×20×20
The two models differ only at layer 10:
• Baseline: C2PSA (attention), 990,976 params
• Proposed: C2TSVMamba, 1,811,201 params
All other layers are identical.
Whole model: 11,434,463 → 12,254,688 params
Neck (FPN + PAN)
P3 (stride 8)
16 C3k2
128×80×80
17 Conv 3×3, s2
128×40×40
15 Concat (+4)
512×80×80
18 Concat (+13)
384×40×40
14 Upsample ×2
256×80×80
19 C3k2
256×40×40
P4
13 C3k2
256×40×40
20 Conv 3×3, s2
256×20×20
12 Concat (+6)
768×40×40
21 Concat (+10)
768×20×20
11 Upsample ×2
512×40×40
22 C3k2 (attn)
512×20×20
P5
Head
23 Segment26
Dual heads:
one-to-many (training)
one-to-one (inference)
Per object:
box, class score,
32 mask coefficients
Proto26:
32 prototypes 160×160
End-to-end inference,
no NMS
layer-10 output feeds layers 11 and 21